Extension I — Sphere Foundations

NOTEBOOK FLOW: read straight down the page, then continue on the next sheet.

LESSON 01 / 05

A sphere is a fixed-distance locus

With centre $C=(h,k,l)$ and radius $R>0$,

$$(x-h)^2+(y-k)^2+(z-l)^2=R^2.$$

Expanding and renaming coefficients gives

$$x^2+y^2+z^2+2ux+2vy+2wz+d=0,$$

with

$$C=(-u,\,-v,\,-w),\qquad R^2=u^2+v^2+w^2-d.$$

The sign of R^2 classifies the locus: positive gives a real sphere, zero a single point, and negative no real points.

WORKED EXAMPLE

Complete squares instead of guessing signs

For

$$x^2+y^2+z^2+2x-4y+6z-11=0,$$

move the constant and complete three squares:

$$(x+1)^2+(y-2)^2+(z-3)^2=25.$$

Thus the centre is $(-1,2,-3)$ and the radius is 5. Equivalently, $(u,v,w)=(1,-2,3)$ and $R^2=1+4+9-(-11)=25$.

The diameter form is a perpendicularity condition

If AB is a diameter, every other point P of the sphere satisfies $\angle APB=90^\circ$. Hence

$$(P-A)\cdot(P-B)=0.$$

The centre is the midpoint of AB and the radius is $AB/2$.

WORKED EXAMPLE

A sphere from two diameter endpoints

For $A=(1,0,0)$ and $B=(0,2,0)$,

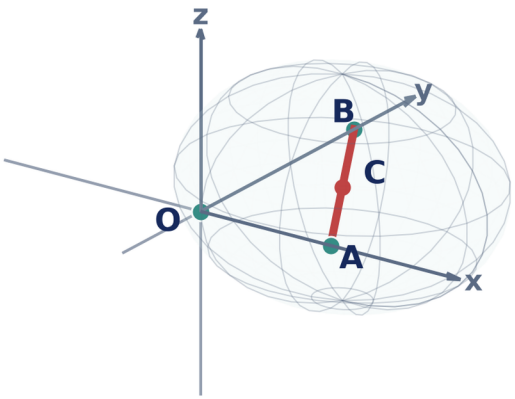
$$C=\left(\tfrac{1}{2},1,0\right), \qquad R=\tfrac{\sqrt{5}}{2}.$$

The equation is

$$\left(x-\tfrac{1}{2}\right)^2+(y-1)^2+z^2=\tfrac{5}{4},$$

or $x^2+y^2+z^2-x-2y=0$. The origin, A and B all satisfy it.

The midpoint of a diameter is the sphere's centre



READ THE FIGURE

C=(1/2,1,0), AB=sqrt(5), and R=sqrt(5)/2. The origin also lies on the sphere because OA is perpendicular to OB.

Four noncoplanar points determine a unique sphere

Substitute the four points into the general equation. Subtracting one equation from the other three eliminates d and leaves three linear equations for the centre coordinates. Noncoplanarity guarantees uniqueness.

WORKED EXAMPLE

A four-point sphere with exact centre and radius

For

$$A=(1,0,0),\quad B=(0,2,0),\quad C=(0,0,3),\quad D=(1,1,1),$$

equal squared distances from $O=(x,y,z)$ give

$$-2x+4y=3,\qquad -2x+6z=8,\qquad y+z=1.$$

Solving,

$$O=\left(-\tfrac{13}{10},\tfrac{1}{10},\tfrac{9}{10}\right).$$

Then

$$R^2=OA^2=\left(\tfrac{23}{10}\right)^2+\left(-\tfrac{1}{10}\right)^2+\left(-\tfrac{9}{10}\right)^2=\tfrac{611}{100}.$$

Thus $R=\sqrt{611}/10\approx 2.4718$. The other three distance checks agree.

COMMON TRAP

Centre signs and the constant term are independent

In the general equation, the centre is $(-u,-v,-w)$, while $d=u^2+v^2+w^2-R^2$. Do not read the centre directly from the linear coefficients without the minus signs, and do not use R^2 itself as the constant d.

KEY RESULT

Second read: four equal squared distances

For the four-point example,

$$|OA|^2=|OB|^2=|OC|^2=|OD|^2=\tfrac{611}{100}.$$

After clearing denominators, the four squared-distance numerators are $529+1+81$, $169+361+81$, $169+1+441$ and $529+81+1$, all equal to 611. Thus the centre and radius are checked without solving the system again.

LESSON 02 / 05

Compare centre–plane distance with the radius

For a sphere of radius R and a plane at distance d from its centre:

- $d < R$: the section is a circle;
- $d = R$: the plane is tangent at one point;
- $d > R$: there is no real intersection.

For a secant plane, the circle's centre Q is the foot of the perpendicular from the sphere's centre C. Pythagoras gives its radius:

$$r^2 = R^2 - d^2.$$

The vector CQ is normal to the plane, while every radius of the section circle lies in the plane.

WORKED EXAMPLE

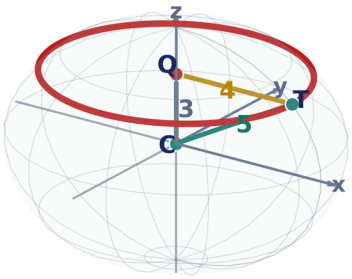
An axis-aligned section makes Pythagoras visible

The plane $z = 3$ meets $x^2 + y^2 + z^2 = 25$ in

$$x^2 + y^2 = 16, \qquad z = 3.$$

Its centre is $(0, 0, 3)$ and radius is 4. The sphere radius 5 is the hypotenuse of a 3-4-5 right triangle.

A plane section is a circle whose centre is the perpendicular foot



READ THE FIGURE

CQ=3 is perpendicular to the section plane, QT=4 lies in it, and CT=5 is a sphere radius. Hence 3^2+4^2=5^2.

WORKED EXAMPLE

A non-axis-aligned section with exact centre

For $2x - 3y + 4z = 20$ and $x^2 + y^2 + z^2 = 25$,

$$d = \frac{20}{\sqrt{29}} < 5.$$

The section centre is the foot from the origin:

$$Q = \tfrac{20}{29}(2, -3, 4) = (\tfrac{40}{29}, -\tfrac{60}{29}, \tfrac{80}{29}).$$

Its radius is

$$r = \sqrt{25 - \tfrac{400}{29}} = \sqrt{\tfrac{325}{29}} \approx 3.3477.$$

The tangent plane is perpendicular to the radius

At a point T on a sphere with centre C, every tangent direction is perpendicular to CT. Therefore the tangent plane is

$$(\mathbf{r} - \mathbf{t}) \cdot (\mathbf{t} - \mathbf{c}) = 0.$$

For $x^2 + y^2 + z^2 = R^2$, this reduces to $\mathbf{t} \cdot \mathbf{r} = R^2$.

WORKED EXAMPLE

Tangency checked by distance

At $T = (3, 0, 4)$ on $x^2 + y^2 + z^2 = 25$, the tangent plane is

$$3x + 4z = 25.$$

The origin is at distance $25/\sqrt{3^2 + 4^2} = 5$ from this plane, exactly the sphere radius. In contrast, $x + 2y + 2z = 9$ has distance $3 < 5$, so it cuts the sphere in a circle rather than touching it.

KEY RESULT

Section centre, section radius and tangent plane are linked

The foot formula supplies the section centre, $r^2 = R^2 - d^2$ supplies its radius, and $d = R$ is the boundary where the circle shrinks to the tangency point. Always compare d with R before declaring an intersection.

LESSON 03 / 05

Subtracting sphere equations leaves a radical plane

The power of P with respect to a sphere with centre C and radius R is $|P-C|^2-R^2$. For two spheres, subtracting their equations cancels all quadratic terms and gives the **radical plane**: the locus of points having equal power with respect to both. It is perpendicular to the line of centres. If the spheres intersect, their intersection circle lies in this plane.

WORKED EXAMPLE

A radical plane with a power check

For

$$x^2+y^2+z^2=25, \qquad (x-3)^2+y^2+z^2=16,$$

write the second as $x^2+y^2+z^2-6x-7=0$. Subtracting gives

$$6x-18=0 \quad\Longrightarrow\quad x=3.$$

At $(3,1,2)$, both powers are -11. The intersection circle is $y^2+z^2=16$ in the plane $x=3$.

Orthogonal spheres have perpendicular radii at intersection

Two intersecting spheres are orthogonal when their radii to a common point are perpendicular. If their centres are distance d apart, the triangle formed by the centres and that point gives

$$d^2=r_1^2+r_2^2.$$

In general-form coefficients this is equivalently

$$2(u_1u_2+v_1v_2+w_1w_2)=d_1+d_2.$$

WORKED EXAMPLE

Orthogonality checked two ways

For $x^2+y^2+z^2-6x=0$ and $x^2+y^2+z^2-6y=0$, the centres are $(3,0,0)$ and $(0,3,0)$, and both radii are 3. Hence

$$d^2=18=9+9=r_1^2+r_2^2.$$

The coefficient form gives $0=0$ as well. In contrast, radii $5,4$ with centre distance 5 do not satisfy orthogonality because $25\neq 25+16$.

Tangents from an external point form a cone

For P outside a sphere, let $D=PC>R$. Every tangent segment has length $\sqrt{D^2-R^2}$. The semi-vertical angle α of the tangent cone satisfies

$$\sin\alpha = \frac{R}{D}.$$

The contact points form a circle in the polar plane, at distance R^2/D from C along CP, with circle radius

$$\rho = R\sqrt{1-\frac{R^2}{D^2}}.$$

WORKED EXAMPLE

The 5–12–13 tangent cone

For $P=(13,0,0)$ and $x^2+y^2+z^2=25$,

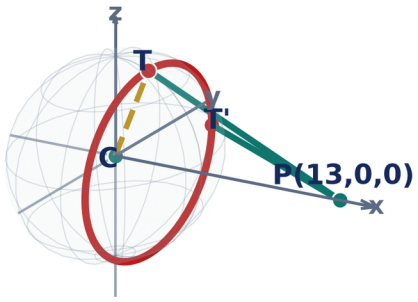
$$D=13, \qquad \sin\alpha=\frac{5}{13}, \qquad \alpha\approx 22.62^\circ.$$

The tangent length is 12. The contact circle lies in

$$x=\frac{25}{13}, \qquad y^2+z^2=\left(\frac{60}{13}\right)^2,$$

so its radius is $60/13$. Its plane is the polar plane $13x=25$ of P.

The contact circle lies in the external point's polar plane



READ THE FIGURE

CT=5 is perpendicular to PT=12, while CP=13. Every contact point lies on x=25/13 and has y^2+z^2=(60/13)^2.

Polar and radical constructions extend naturally

For $x^2+y^2+z^2=R^2$, the polar plane of $P=(x_1,y_1,z_1)$ is

$$xx_1+yy_1+zz_1=R^2.$$

For $P=(1,2,3)$ and $R=5$, it is $x+2y+3z=25$. Since P is inside the sphere, no real tangent cone exists, but the polar plane remains defined. If the polar of P contains Q, the polar of Q contains P.

WORKED EXAMPLE

Reciprocity checked on a contact point

For $P=(13,0,0)$ and $R=5$, take the contact point $T=(25/13,0,60/13)$. The polar of P is $13x=25$, which contains T. Conversely, the polar of T is

$$\tfrac{25}{13}x+\tfrac{60}{13}z=25,$$

and substituting P gives $25=25$. This verifies the reciprocal polar relation using the tangent construction.

Three spheres generally have three radical planes sharing a **radical axis**. For the source spheres $x^2+y^2+z^2=14$, $(x-5)^2+y^2+z^2=29$ and $x^2+(y-5)^2+z^2=19$, those planes give $x=1$, $y=2$ and $y=x+1$. Thus the radical axis is $(1,2,t)$, and every point on it has common power t^2-9.

KEY RESULT

A useful coaxal extension

A sphere orthogonal to two given spheres is orthogonal to every member of their coaxal pencil, and its centre lies on their radical plane. This follows by linearity of the general-form orthogonality condition.

LESSON 04 / 05

Volume and surface area determine the inradius

Join the incentre of a tetrahedron to its four vertices. This decomposes the solid into four pyramids whose heights are the inradius r. If V is the volume and S the total surface area,

$$V = \tfrac{1}{3} r S, \qquad r = \tfrac{3V}{S}.$$

With one vertex at the origin and the others at $\mathbf{a}, \mathbf{b}, \mathbf{c}$,

$$V = \tfrac{1}{6} |[\mathbf{a}\ \mathbf{b}\ \mathbf{c}]|,$$

and four cross products give the face areas.

WORKED EXAMPLE

An exact inradius and incentre

For the tetrahedron

$$O = (0,0,0), \quad A = (4,0,0), \quad B = (0,4,0), \quad C = (0,0,3),$$

the coordinate-face areas are $8, 6, 6$, and the slant face has area $2\sqrt{34}$. Thus

$$S = 20 + 2\sqrt{34}, \qquad V = \tfrac{4 \cdot 4 \cdot 3}{6} = 8,$$

so

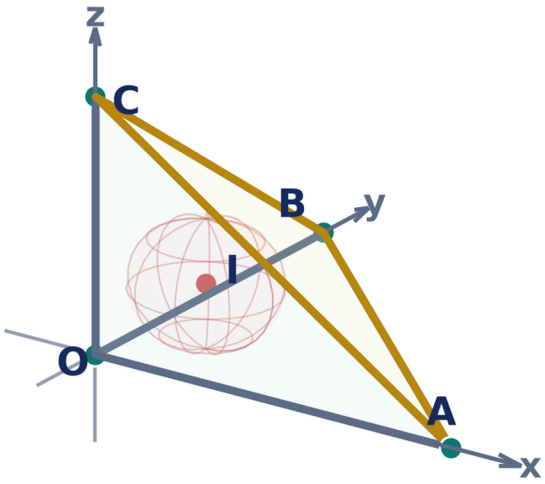
$$r = \frac{24}{20 + 2\sqrt{34}} = \frac{12}{10 + \sqrt{34}} \approx 0.7580.$$

The incentre is $I = (r, r, r)$: its distances to the three coordinate planes are r. The slant plane is $3x + 3y + 4z = 12$, and

$$\frac{12 - 10r}{\sqrt{34}} = r$$

is exactly the defining equation $12 = (10 + \sqrt{34})r$.

The insphere is tangent to all four faces



READ THE FIGURE
The rounded plotted radius is r=0.758; the exact value is 12/(10+sqrt(34)). Its centre I=(r,r,r) is equidistant from all four faces.

Monge's six midplanes concur

For each edge AB of a tetrahedron, form the plane through the midpoint of AB and perpendicular to the opposite edge CD. These six **Monge planes** are concurrent.

Place the circumcentre at the origin, so $|\mathbf{a}| = |\mathbf{b}| = |\mathbf{c}| = |\mathbf{d}| = R$, and define

$$\mathbf{m} = \frac{\mathbf{a}+\mathbf{b}+\mathbf{c}+\mathbf{d}}{2}.$$

For the midplane associated with AB,

$$\left(\mathbf{m} - \frac{\mathbf{a}+\mathbf{b}}{2}\right)\cdot(\mathbf{d}-\mathbf{c}) = \frac{\mathbf{c}+\mathbf{d}}{2}\cdot(\mathbf{d}-\mathbf{c}) = \frac{|\mathbf{d}|^2 - |\mathbf{c}|^2}{2} = 0.$$

Cyclic relabelling proves all six concurrences. In general coordinates,

$$\mathbf{m} = 2\mathbf{g} - \mathbf{o},$$

where $\mathbf{g}$ is the centroid and $\mathbf{o}$ the circumcentre. Thus the Monge point is the reflection of the circumcentre in the centroid.

WORKED EXAMPLE

An exact Monge point for a nonregular tetrahedron

For

$$A=(1,0,0),\quad B=(0,2,0),\quad C=(0,0,3),\quad D=(2,1,4),$$

solving equal distances to the four vertices gives

$$O=\left(\tfrac{47}{34},\tfrac{49}{34},\tfrac{61}{34}\right).$$

The centroid is $G=(3/4,3/4,7/4)$, so

$$M=2G-O=\left(\tfrac{2}{17},\tfrac{1}{17},\tfrac{29}{17}\right).$$

Substitution into all six midplane equations confirms the concurrence exactly.

SHORT DERIVATION

One of the six midplane checks

For edge AB, the midpoint is $(1/2,1,0)$ and the opposite direction is $D-C=(2,1,1)$. Hence

$$M-\left(\tfrac{1}{2},1,0\right)=\left(-\tfrac{13}{34},-\tfrac{16}{17},\tfrac{29}{17}\right).$$

Its dot product with $(2,1,1)$ is $(-26-32+58)/34=0$. The other five checks use the same perpendicularity condition.

LESSON 05 / 05

Spherical excess measures area

A spherical triangle is bounded by arcs of three great circles. Its angles are the angles between the corresponding great-circle planes. On a sphere of radius R, Girard's theorem gives

$$E = A + B + C - \pi > 0, \qquad \text{area} = R^2 E.$$

Angles must be in radians in this formula. The fraction of the whole sphere covered is $E/(4\pi)$.

WORKED EXAMPLE

Area from three spherical angles

On a unit sphere, angles $80^\circ, 70^\circ, 60^\circ$ give

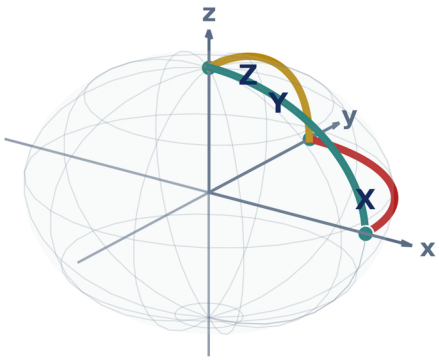
$$E = (80 + 70 + 60 - 180)^\circ = 30^\circ = \frac{\pi}{6}.$$

Thus the area is $\pi/6 \approx 0.5236$, and the fraction of the sphere is

$$\frac{\pi/6}{4\pi} = \frac{1}{24}.$$

If a sphere of radius 2 has spherical-triangle area 1, its excess is $1/4$ radian.

One octant is a spherical triangle with three right angles



READ THE FIGURE

The three arcs lie in perpendicular coordinate planes. E=3(pi/2)-pi=pi/2, so the area is pi/2, exactly one eighth of 4 pi.

A regular tetrahedron from symmetric coordinates

Take

$$A = (1, 1, 1), \quad B = (1, -1, -1), \quad C = (-1, 1, -1), \quad D = (-1, -1, 1).$$

Every edge has length $2\sqrt{2}$. The centroid and circumcentre are the origin, so $R=\sqrt{3}$. The face BCD lies in $x+y+z=-1$, whose distance from the origin is $1/\sqrt{3}$; hence $r=1/\sqrt{3}$ and $R=3r$.

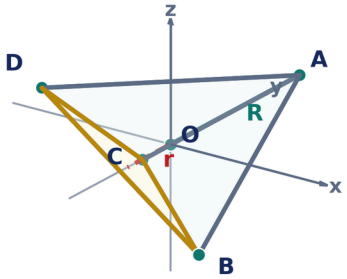
Scaling to edge length a gives

$$R=\frac{a\sqrt{6}}{4},\qquad r=\frac{a\sqrt{6}}{12},\qquad V=\frac{a^3}{6\sqrt{2}}.$$

The internal dihedral angle between any two faces is

$$\theta=\cos^{-1}\left(\tfrac{1}{3}\right)\approx 70.53^\circ.$$

In a regular tetrahedron, the circumradius is three times the inradius



READ THE FIGURE
For edge 2 sqrt(2), OA=R=sqrt(3). The foot to BCD is its centroid (-1/3,-1/3,-1/3), at distance r=1/sqrt(3).

WORKED EXAMPLE

Recover every measurement from the inradius

If $r=1$, then

$$a=\frac{12}{\sqrt{6}}=2\sqrt{6},\qquad R=3,$$

and

$$V=\frac{(2\sqrt{6})^3}{6\sqrt{2}}=8\sqrt{3}\approx 13.8564.$$

The solid angle uses dihedral angles, not face angles

At a vertex, a small sphere cuts out a spherical triangle whose **angles** are the tetrahedron's dihedral angles θ, while its **sides** are the face angles $\pi/3$. Girard's theorem therefore gives the solid angle

$$\Omega=3\theta-\pi=3\cos^{-1}\left(\tfrac{1}{3}\right)-\pi\approx 0.5513\ \text{sr}.$$